2026 PDS A

GitHub repository: `https://github.com/Gags50/2026-PDS-A`

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1 Introduction

1.1 Dataset in general

The dataset was published in October 2020 to combat the lack of public clinical pictures of skin lesions, since CAD systems only worked on dermoscopy images.

The dataset contains images collected from smartphones from 1373 patients with 1641 skin lesions, Consisting of 3 skin diseases:

- (Actinic Keratosis, Nevus and Bowens disease classified under Nevus)

and 3 skin cancers:

- (Basal Cell Carcinoma, Squamous Cell Carcinoma, Melanoma)

The population is importantly biased, with the population largely being picked from people living in brazil, a tropical region. A sizable portion of the patient population are immigrants who're unacclimatized to the climate, and have increased risk of getting skin lesions. While this is not bad, it is important to keep in mind that the unique circumstances of the population may not reflect the global patient population, nor is it certain the data would be equally effective when applied to a different population.

Notably, just 58.8% of the lesion diagnoses have been confirmed through biopsy. 100% of cancerous lesions are biopsied, but only 20,9% of the skin diseases have been confirmed through biopsy. This could potentially show a schism between the two types of lesions when it comes to the reliability of real life application of the data.

The pictures are also all raw, unprocessed images. Some processing has been shown to enhance the efficacy of related models, and could be a valuable consideration in the future handling of the data.

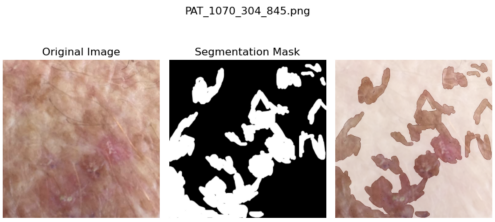
The fact that the pictures are taken with different types of smartphone cameras can also cause some unintended and hard to detect biases in the model. The model may pick up on the small artifacts or quirks in how the individual cameras produce images and eventually develop rules based on those, in spite of their irrelevance to the actual diagnosis.

1.1.1 Analysis of lesion and masks:

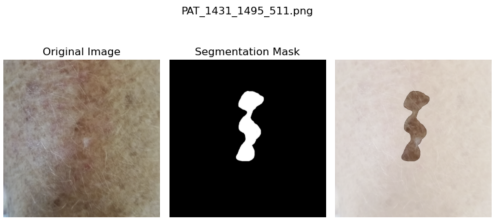
- The outline of some mask don't correspond with the lesion on the original image.
  See figure: 1a
- Masks don't cover the whole lesion, they cut off certain parts if the lesion is not nicely shaped.
  See figure: 1c
- Oddly colored skin and other colored marks can confuse the mask.
  See figure: 1a
- What does the masks include, can they be influenced by other factors than the skin lesion. Such as very wrinkly, hairy or dry skin.
  See figure: 1a
- Pictures are not always only of skin with a lesion, sometimes there are things in the background.
  See figure: 1b
- Because pictures are taken on smartphones, resolution, lighting quality and blurriness may all impact the fidelity of the picture.
  See figure: 1b

- Pen marks are different shapes - circles - dots or asymmetrical shapes.
- Different depths can confuse - unclear image vs mask

1.1.2 Pictures of lesion, mask and overlay:

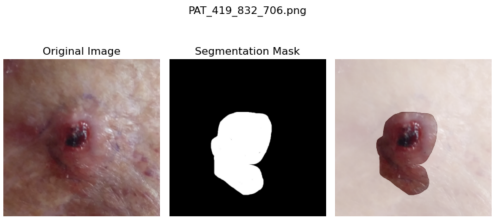


(a) PAT_1070_304_845



(b) PAT_1431_1495_511

(c) PAT_1701_3135_5



(d) PAT_419_832_706